Computer Communication Networks

Data Link Layer: CRC and Checksum

Session 9

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Data Link Layer

Learning Outcome:- At the end of the lesson, you are expected to be able to:

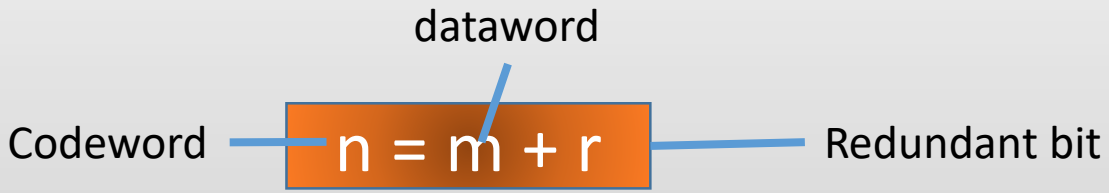
- Summary of VRC, LRC, and 2-D Parity Check
- Discuss CRC examples
- Demonstrate Checksum.



Data Link Layer

Error Control

- Two basic strategies of implementing error control at DDL:
  1. Error Detection
  2. Error Correction
- To implement error control, consider the following:
  1. Message bit = ***m-data***
  2. Check bits = ***r-redundant bits***
  - The total number of bits is now ***n = m + r.***
- An n-bit unit consisting of both **data** and check bit is called n-bit ***codeword***





Data Link Layer

Error Control

- Error detection involves checking of the received data for any errors.
- The number of errors or the type of errors does not have an impact.
- If errors are detected, the receiver requests for retransmission.
- Three main techniques of error detection are
  1. Parity Check
  2. CRC
  3. Checksum

1. VRC
2. LRC
3. 2-D Parity Check



Data Link Layer

Error Control

- Vertical Redundancy Check (VRC)
  - A simple and least expensive mechanism for error detection is the simple parity check.
  - It is also known as One-Dimensional Parity Check.
  - One redundant bit called as parity bit is appended to every data unit such that the total number of 1's in the data unit including the parity bit becomes even.

Data Word	Parity Bit	Code Word
101001	1	1010011
000	0	0000
100100	0	1001000
1111001	1	11110011



Data Link Layer

Error Control

- Vertical Redundancy Check (VRC)
  - VRC can detect all single bit errors that happen in each **code word**.
  - However, the problem with VRC is that it can detect only odd number of errors
    - ✓ Code word sent = 0000
    - ✓ After transmission receiver received = 0110
      - Receiver counts the number of 1's and assumes there is no error.
      - This is the draw back of VRC



Error Control

- Longitudinal Redundancy Check (LRC)
  - The data to be transmitted is organized into a table of rows and columns.
  - A parity bit is computed for each column and the full block of redundant bits is now appended to the original data and sent to the receiver.
  - The advantage of using LRC over VRC is that it can check all the burst errors.
    - ✓ An LRC of n-bits can detect a burst error of n-bits
  - It cannot detect errors where there are even numbers of damaged bits that are located in the same column position of different rows



Error Control

- 2-D Parity Check
  - The data to be transmitted is organized into a table of rows and columns.
  - A parity bit is computed both rows and columns and the full block of redundant bits is now appended to the original data and sent to the receiver.
  - 2-D Parity Check is better than VRC and LRC
  - It increases the likelihood of detecting burst errors.
    - ✓ A burst error of more than n-bits is also detected.
  - If two bits in one data unit are damaged and two bits in exactly same position in another data unit are also damaged, the 2-D parity check checker will not detect an error.



Error Control

- Example
  1. Suppose the following data blocks are received where 2D parity is used: 10101001 00011001 11011101 11100111 10101010. Check if it has any errors.
  2. The serial port on a computer is set to transmit a file using 8 data bits followed by 1 parity bit. The port is set for even parity. The other computer receives the following 9-bit strings. Determine which are valid and which are not.
     a) 001011010 ✓
     b) 111111111 ✗
     c) 101010100 ✓
  - Find the LRC for the sending data blocks 11100111 11011101 00111001 10101001 and determine the data that is transmitted?



Data Link Layer: Error Detection

❖ Cyclic Redundancy Check:

- It is most powerful and easy to implement technique.
- **CRC is based on binary division**.
- In CRC, a sequence of redundant bits called CRC bits, are appended at the end of data unit so that the resultant data unit is exactly divisible by the some predetermined binary number.
- **If k bit message is to be transmitted:**
  - ✓ transmitter generates r-bit sequence known as frame check sequence(FCS).
  - ✓ So k+r bit data are actually transmitted.
- **How r-bit is generated?**
  - ✓ r-bit is a remainder of k+r
  - ✓ predetermined divisor of r+1 length.



Data Link Layer: CRC

❖ CRC: Cyclic Redundancy Check

- CRC is based on a notion of polynomial codes:
  - A **k-bit** stream is considered as the list of coefficients for a polynomial (*of degree k-1*) with **k** term, ranging from x^{k-1} to x^0
    - ✓ The left most bit represents the coefficient x^{k-1} , the next bit represents the coefficient x^{k-2} and so on… the rightmost bit represents coefficient $x^{0.}$
  - Example: **110101** represents a 6 term polynomial $x^5 + x^4 + x^2 + 1$



Data Link Layer: Error Detection

CRC: Sender

- Algebraic polynomial(dataword) is $x^3 + 1$
- Divisor is $x^3 + x + 1$

Step1:- Convert algebraic polynomial equation to binary number

Step 2:- Convert Divisor polynomial equation to binary number

Step 3:- count the number of bits in divisor and add one less zero bits to dataword

Step 4: Perform binary division (XOR) and find remainder.

Step 5:- Add remainder to dataword and sent the message.



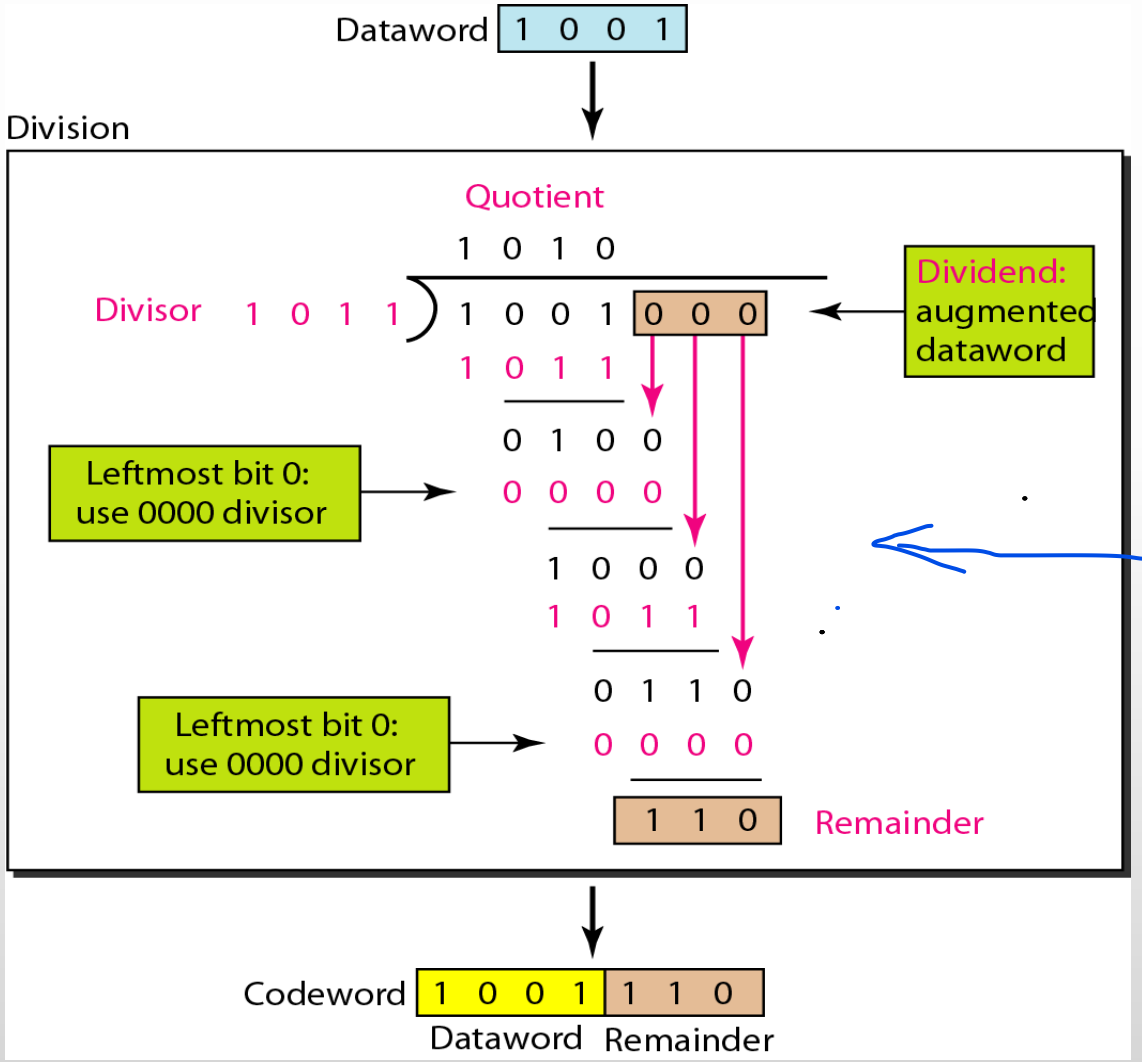
Data Link Layer: Error Detection

❖ CRC: CRC Encoder

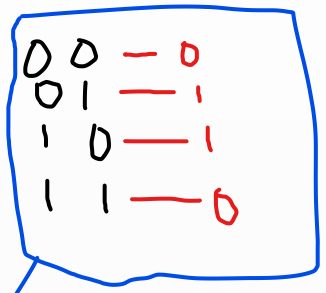
- **Dataword** is 1001 and **divisor** is 1011. What will be the **codeword**?
  - The binary coefficient for dataword **1001000** after augmenting three zeros is $x^6 + x^3$
  - The binary coefficient for divisor **1011** is $x^3 + x + 1$

Data Link Layer: Error Detection

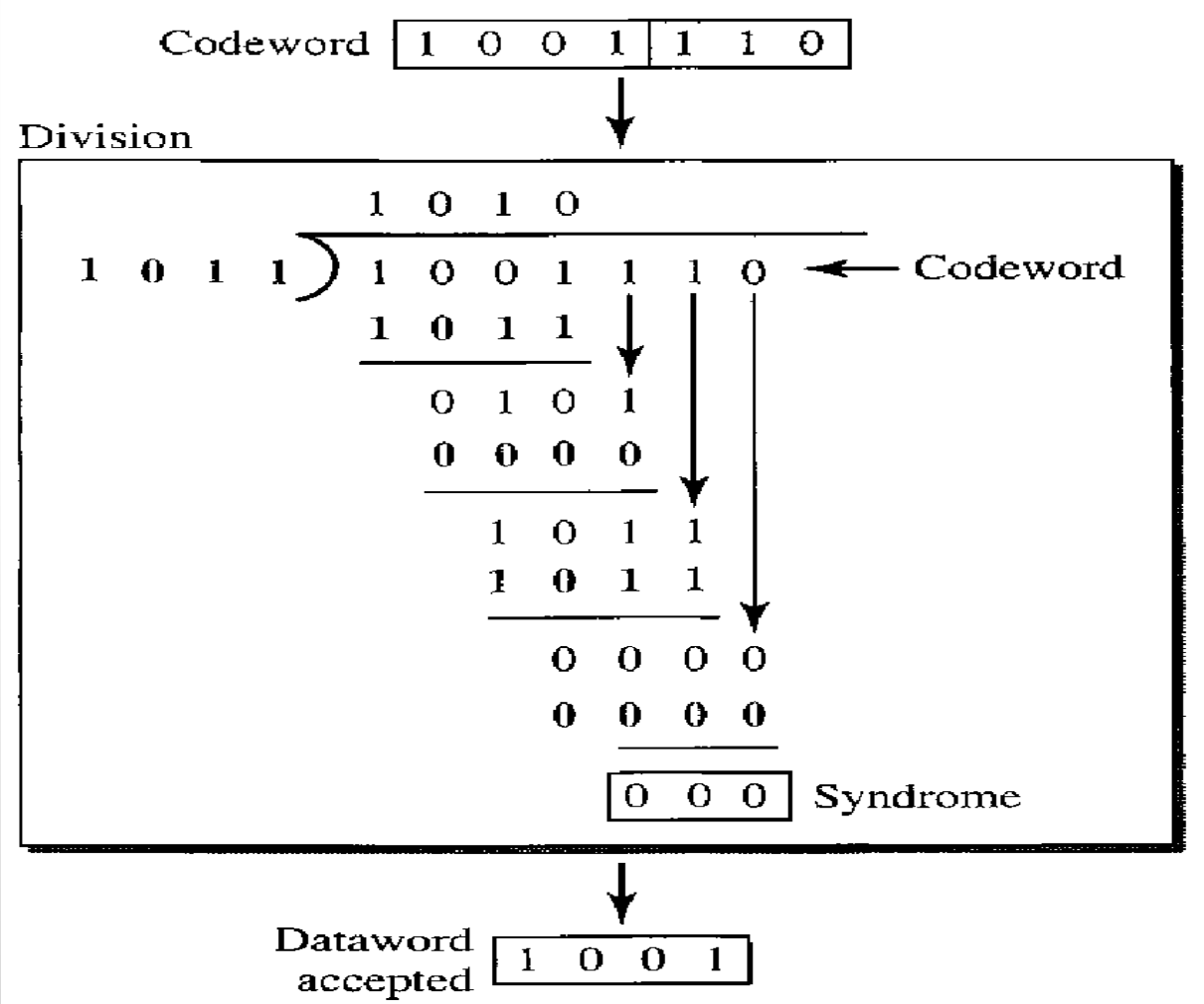
CRC: Sender



XOR XOR opposite of and n or



CRC: Receiver



Reminder = 0 then the info given is correct



Data Link Layer: Error Detection

❖ CRC:

Q. **Given the dataword 1010011110 and the divisor 10111:**

a) Show the generation of the codeword at the sender site(using binary division)
b) Show the checking of the codeword at the receiver site(assume no error)
c) Show the checking of the codeword at the receiver site if codeword received is 10100111101111

Q. **Given the dataword polynomial equation is $x^7+x^6+x^4+x^3+x+1$ and the divisor polynomial equation is x^4+x+1:**

a) Show the generation of the codeword at the sender site(using polynomial division)
b) Show the checking of the codeword at the receiver site(assume no error)

➢ Perform both CRC Encoder and decoder:

1. **Dataword** is 1010 and **divisor** is 10111. What will be the **codeword**?
2. **Dataword** is 1000 and **divisor** is 1011. What will be the **codeword**?



Data Link Layer: Error Detection

- **Home work**

Q. The message 11001001 is to be transmitted using the CRC polynomial x3 + 1 to protect it from errors. What message should be transmitted? If the received codeword at the receiver is 11011001011 then how the receiver will take decision?

Answer: 11001001011

Received CRC 010 is not equal to 011 therefore, reject and message and request sender to retransmit the message.

Q. In CRC, dataword is 01010001 and divisor is 1001. What is the CRC and the codeword that sender will used to transmit the message?

Answer: 010 is the CRC and Codeword is 0101001010



Data Link Layer: Error Detection

❖ CRC:

Q. A bit stream 10011101 is transmitted using standard CRC method. The generator polynomial is $x^3 + 1$. Show actual bit string transmitted. Suppose that the third bit from left is inverted during transmission. Show that this error is detected at the receiver's end.

Solution:- ????



Data Link Layer: Error Detection

Solution:

The frame is 10011101.

The generator is 1001.

The message after appending three zeros is 10011101000.

The remainder on dividing 10011101000 by 1001 is 100.

So, the actual bit string transmitted is 10011101100.

The received bit stream with an error in the third bit from the left is 10111101100. Dividing this by 1001 produces a remainder of 100, which is different from 0.

Thus, the receiver detects the error and can ask for a retransmission



Data Link Layer: Checksum

Checksum

- The Checksum is one of the error detection methods. The checksum is used in the Internet by several protocols.
  - Suppose our data is a list of five 4-bit numbers that we want to send to a destination. In addition to sending these numbers, we send the sum of the numbers.
    - For example, if the set of numbers is (7, 11, 12, 0, 6), we send (7, 11, 12, 0, 6, **36**), where 36 is the sum of the original numbers. The receiver adds the five numbers and compares the result with the sum.
    - If the two are the same, the receiver assumes no error, accepts the five numbers, and discards the sum.
    - Otherwise, there is an error somewhere and the data are not accepted.



Data Link Layer: Checksum

❖ Checksum

- The encoded data 12,23,45,15,29 are to be transmitted using checksum method.
  1. Show actual data to be transmitted?
  2. Show that no error is being detected and accept the data.
  3. Suppose that the last number sum if 66, show error is detected at the receiver's end.



Data Link Layer: Checksum

Checksum

- We can make the job of the receiver easier if we send the negative (complement) of the sum, called the checksum.
- In this case, we send (7, 11, 12, 0, 6, −36).
  - The receiver can add all the numbers received (including the checksum).
    - ✓ If the result is 0, it assumes no error;
    - ✓ otherwise, there is an error.



Checksum

Example: Sender

- Data 10101010 00000000 10001000 10001001

Data 1 0 1 0 1 0 1 0

0 0 0 0 0 0 0 0

1 0 0 0 1 0 0 0

1 0 0 0 1 0 0 1

Sum 1 1 0 1 1 1 0 1 1

1

1 0 1 1 1 1 0 0 Wrap sum

0 1 0 0 0 0 1 1 checksum (*compliment of wrap sum*)

1. Arrange data in tabular form
2. Find the sum→ XOR or Binary addition
3. Wrap the left most carry bit and calculate the sum
4. Perform compliment of wrap sum to get checksum
5. Sent Data + checksum to the receiver.



Checksum

❖ Example: Receiver

➢ Data + Checksum: 10101010 00000000 10001000 10001001 0 1 0 0 0 0 1 1

Data 1 0 1 0 1 0 1 0

0 0 0 0 0 0 0 0

1 0 0 0 1 0 0 0

1 0 0 0 1 0 0 1

0 1 0 0 0 0 1 1

---

1 1 1 1 1 1 1 1 0 sum

1

---

1 1 1 1 1 1 1 1 wrap sum

0 0 0 0 0 0 0 0 Result (*no error detected*)

1. Arrange data + Checksum in tabular form
2. Find the sum
3. Wrap the left most carry bit and calculate the sum
4. Perform compliment of wrap sum
5. Result
   →if Compliment of wrap sum is zero, than no error is detected
   →if compliment of wrap sum is one, than error is detected.



Recapitulation

❖ Example

Q. A sender needs to send the four data items 11010110 10101100 10111001 10010011. Answer the following:

1. Find the internet checksum at the sender site
2. Find the actual data for the transmission to the receivers.
3. Calculate the error if third data is change to 10101010 while transmitting via unreliable link.



Recapitulation

❖ Example

Q. Suppose that a message 1001 1100 1010 0011 is transmitted using Internet checksum (4 bits word). What is the value of checksum?



Recapitulation

- **We have learned:**
  - Seem more examples on CRC
  - Discussed checksum



References

References:

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4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan